Ethan Feng

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Professional Summary

Computational biologist and AI engineer with hands-on experience building RAG systems for healthcare at NASA and RediMinds - including a hybrid graph-vector system recognized in NASA's New Technology Report. Research background spans medical imaging, ML fairness, and biomedical informatics. Looking to apply both the biology intuition and the engineering depth to impactful problems in health-tech or life sciences AI.

EDUCATION

Washington University in St. Louis — **Aug 2022 - May 2026**
B.S. in Computational Biology, Minor in Computer Science — *St. Louis, MO*
- **GPA:** 3.9/4.0
- **Achievements:** College of Arts and Sciences Dean's List (8/8 semesters)
- **Coursework:** Machine Learning, Practical Bioinformatics, Data Structures & Algorithms, Object-Oriented Programming, Intro Computer Engineering, Prob/Stats, Population Genetics, Matrix Algebra, Biology, Organic Chemistry, Physics

EXPERIENCE

RediMinds Inc. — **Jan 2026 - Present**
AI Engineer — *Detroit, MI / remote*
- Lead development of an automated workers' compensation medical necessity review system using hierarchical RAG and agent frameworks (Python, PostgreSQL, Google Cloud SDK), reducing review time by ~95% compared to manual workflows
- Rapidly prototyped AI features into production workflow software to solve client pain points - including agentic vectorless RAG, OCR, automated extraction, and audio transcription, reducing client workflow times by 85%

National Aeronautics & Space Administration (NASA) — **Jun 2025 - Aug 2025**
AI Systems Engineering Intern — *Houston, TX*
- Devised a hybrid graph-plus-vector retrieval-augmented generation (RAG) AI system with PyTorch and LangGraph to help Biomedical Flight Controllers perform daily tasks and troubleshoot ISS crew health issues, enabling faster decision-making
- Increased information retrieval efficiency by optimizing indexing and query pipelines; the improvement was highlighted in NASA's New Technology Report (MSC-28048) and added to the JSC Software Portfolio
- Developed search-engine algorithms and embedding-based semantic search to explore large astronaut health data sets, combining NLP and large language models for better data retrieval; built an intuitive GUI with Streamlit that shortens the learning curve for decision makers

Washington University in St. Louis, Institute for Biomedical Informatics — **Aug 2024 - Jan 2025**
Medical Imaging Al Undergraduate Research Intern — *St. Louis, MO*
- Led an independent study evaluating SAM2 across multiple biomedical imaging modalities, characterizing performance boundaries and generalization limits to assess readiness for clinical deployment
- Investigated computer vision technology to improve performance and fairness in medical AI, applying Python image processing techniques

Vanderbilt University, Department of Biomedical Informatics — **May 2024 - Aug 2024**
Machine Learning Research Intern — *Nashville, TN*
- Investigated how synthetic dermatology medical images could augment training data sets to improve fairness between skin tones in machine learning classification models
- Discovered potential for GAN-generated synthetic data to increase both AUC and fairness without tradeoffs

Washington University in St. Louis, Sah Lab — **May 2023 - Nov 2023**
Biology Undergraduate Vagelos Research Fellow — *St. Louis, MO*
- Investigated the role of anion channel SWELL1 in the recruitment of satellite muscle cells in response to injury
- Performed Western blot, skeletal muscle tissue sectioning, and microscopy to analyze protein expression and tissue morphology, to characterize SWELL1's role in satellite cell recruitment following skeletal muscle injury
- Participated in the Vagelos Fellowship program, conducting fully funded summer research that advanced knowledge in muscle cell recruitment and repair

Washington University in St. Louis, Varsity Swim Team — **Aug 2022 - May 2026**
Student Athlete — *St. Louis, MO*
- Managed 22+ hours of weekly training while maintaining high academic performance, demonstrating strong time management and resilience
- Achieved 5x NCAA All-American status and became an NCAA finalist, earning recognition as a CSCAA Individual Scholar All-American

SKILLS

- **Languages (ordered by proficiency)**: Python, Java, Bash, C, C++, Swift, R
- **Tools and Frameworks**: Claude Code, OpenClaw, LangGraph, Google Agent Development Kit, PostgreSQL, Google Cloud Platform, Neo4j, Pytorch, Tensorflow, Git, Streamlit, Vector/Graph RAGS, GANs, embedding-based semantic search, CNNs, Segment Anything 2 (SAM2), Model Context Protocol, Al agent design & prototyping, LLMs, Cursor, Agentic RAG